SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

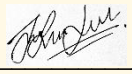
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 1– Pseudocode
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
Student Name:	Joel Benn John B	**Reg. No.:**	1911260003
Section / Batch:	Slot A	**Date:**	23-10-26

Code of Conduct

I Joel Benn John B (1911260003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Duration: 1 hour
- Answer all Questions
- Write Pseudocode in paper and upload program in GITHUB

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. Design a health and self-care mobile app that helps users maintain mental wellbeing with meditations and exercises. This case study talks you through the design process of creating a user- friendly mobile app. Design UI with the following requirements.
   - User Login
   - Register
   - Health Routine Blogs
   - Self-Care Blogs

2. This is an intelligent job assistant that helps users find their ideal job and get to all the way to signing a contract faster and more easily. Design UI with the following requirements.
   - User Login
   - Register Subscription
   - Personal Detail
   - Resume Upload
   - Choosing Job

3. This is a mobile app that can help people get parking slots easily even when they travel beyond their normal routes. Design UI with the following requirements.
   - UserLogin
   - Register Count of Empty Parking
   - Free Parking Slots Display
   - Booking Parking Slots

4. Design a mobile app for Online Body Fitness tips and videos for a good daily routine. Contains the following requirements.

   User Login
   - Register
   - Fitness Blog
   - Diet Planner
   - Fitness Videos
   - Payment and Subscription

NAME: JOEL BENN JOHNS	COURSE: MOBILE COMPUTING
R.NO: 1911260003	COURSE CODE: ITA 0313 - SLOT A
DATE: 23-10-26	CO3 - AT1 - PSEUDOCODE.

1. Health & Self - Care App.

```
START Health App

// Authentication.
SHOW screen - Login
If User clicks "Register" THEN
    SHOW Screen - Register
    COLLECT Name, Email, Password → SAVE Account
END IF

// Main Navigation & content
SHOW Screen - Home

WHILE User active DO
    IF Selected "Blogs" THEN
        SHOW Screen - BlogFeed
        SWITCH Filter:
            CASE "Health Routine": Display Health - Articles
            CASE "Self - care": Display Self care - Articles
        END SWITCH
        IF Clicked Article THEN SHOW Screen - Blog Detail
    END IF
END While
END HealthApp.
```

2. Intelligent Job Assistant App.

```
START JobAssistantApp

// onboarding & Setup.
SHOW Screen_Login / Screen_Register.
SHOW Screen_Subscriptions (Select Free or
Premium Tier)
SHOW Screen_Personal Details (Input Name, Role,
Location)
SHOW Screen_Resume Upload (Upload PDF → Extract
Skills)

// Matching & Job Selection.
SHOW Screen_Jobchoosing
MATCH Jobs WHERE Job.Requirements == User.Skills

FOR EACH Matched_Job DO
    DISPLAY Job_Card (Title, Company, Match%)
END FOR

IF User clicks "Apply / Sign Contract" THEN
    SHOW Application_Success_Modal

END IF

END JobAssistantApp
```

3. Parking Slot Finder App.

```
START ParkingFinderApp

// Access & Dashboard
SHOW Screen – Login
SHOW Screen – Dashboard
FETCH Live – Parking – Data

// Display Count & Empty Slots
total – empty = COUNT (Slots WHERE status == "Available")
DISPLAY total – empty ON Screen – Dashboard.

IF User Searches destination THEN
    SHOW Screen – Slot Display (List empty slots,
rates, distance)
END IF
// Booking Process
IF User selects slot THEN
    SHOW Screen – Booking
    INPUT Vehicle – Number, Duration
    Calculate Total – Cost = Duration * Rate

    IF clicked "Confirm Booking" THEN
        UPDATE Slot. Status = "Reserved"
        GENERATE Digital – Ticket

    END IF
END IF

END ParkingFinderApp
```

4. Online Body Fitness App.

```
START Fitness App

// Auth & Access
SHOW screen - Auth (Login / Register)

// Feature Navigation
SHOW screen - Dashboard

WHILE APP open DO
    SWITCH User_Selection :
        CASE "Fitness Blog" :
            SHOW screen - FitnessBlog (Read workout tips)
        CASE "Diet Planner" :
            SHOW screen - DietPlanner (Input goals -> Generate meal plan)
        CASE "Fitness Videos" :
            SHOW screen - VideoLibrary (Play workout videos)
        CASE "Subscription" :
            SHOW screen - payment
            INPUT Payment - Details -> Unlock Premium - Access
    END SWITCH
END WHILE

END Fitness App
```